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Unveiling Patterns: Adaptive Feature Selection and Sequential Clustering of Nanopore Signals: Midterm Review

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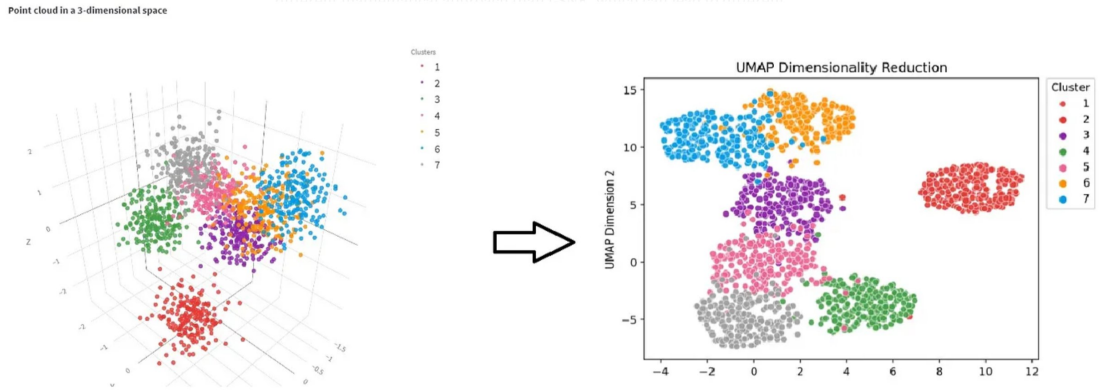
Goal

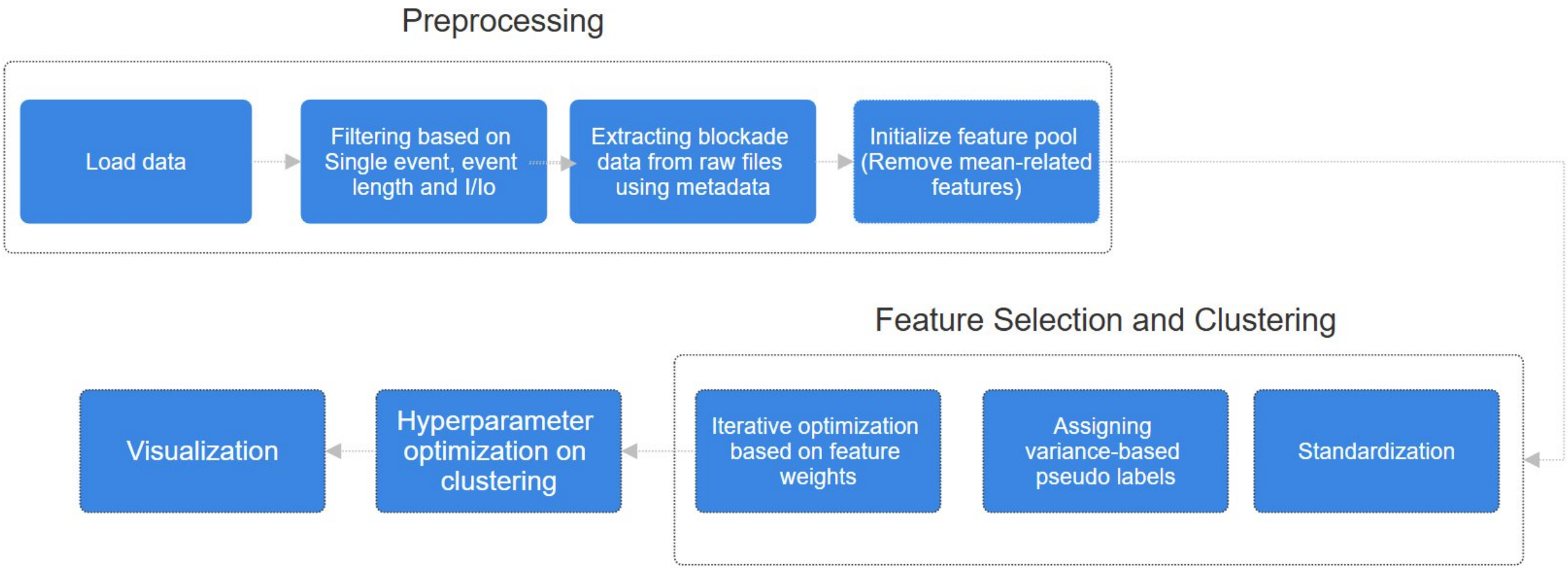
Developing an adaptive feature-based clustering and pattern detection approach

- **The aim is to integrate feature selection directly into the clustering algorithm,** leveraging existing clustering algorithms
- **We introduce analytes dynamically**
- **We apply visualization techniques like UMAP**



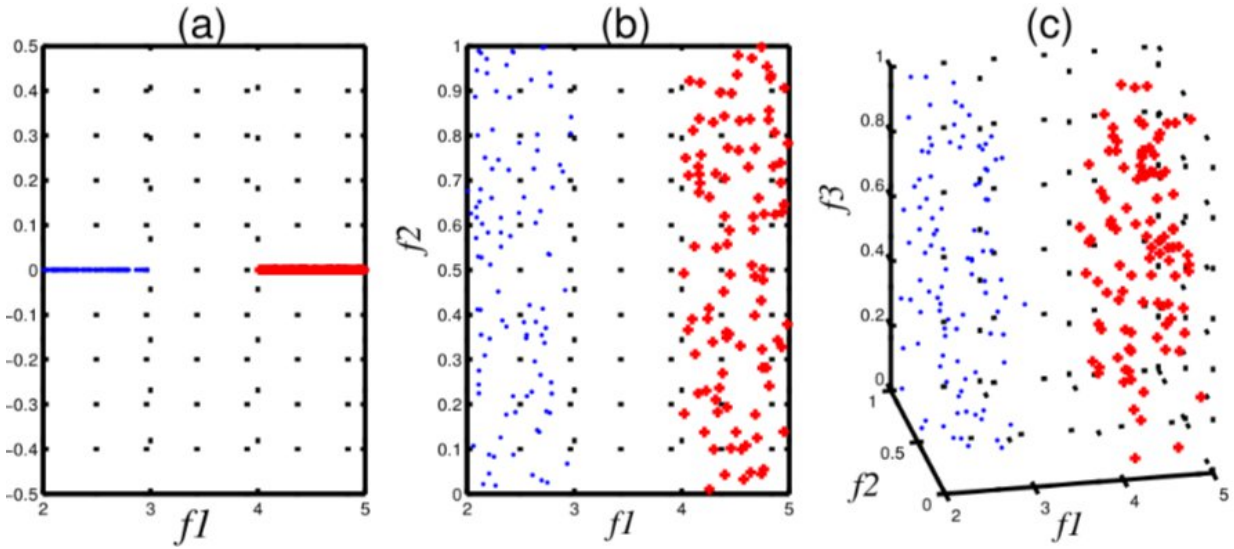
UMAP visualization [2]

Pipeline


Preprocessing
Load data
Filtering based on Single event, event length and I/Io
Extracting blockade data from raw files using metadata
Initialize feature pool (Remove mean-related features)
Feature Selection and Clustering
Standardization
Assigning variance-based pseudo labels
Iterative optimization based on feature weights
Hyperparameter optimization on clustering
Visualization

Why Feature Selection is Necessary

- More features != More information



[2]

Lasso Sparse Feature Selection [1]

What is Lasso?

- Regression method that performs both *variable selection* and *regularization*
- Penalizes absolute size of regression coefficients, encouraging sparsity

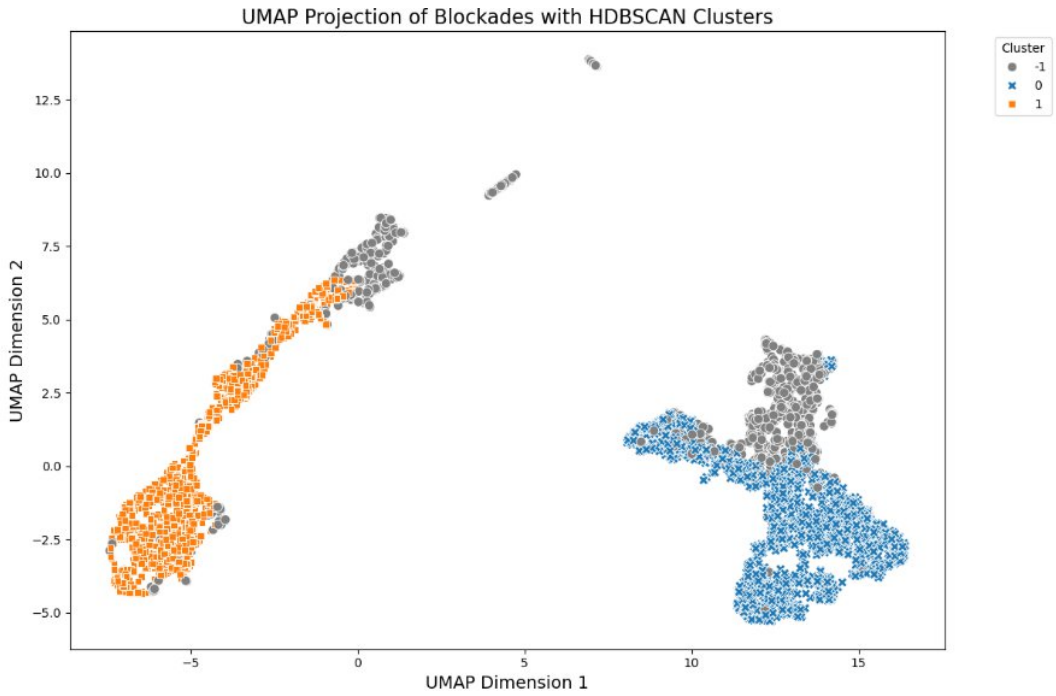
Key Features

- Uses *L1 regularization*, which adds a penalty proportional to the absolute value of the coefficients
- Helps to avoid overfitting

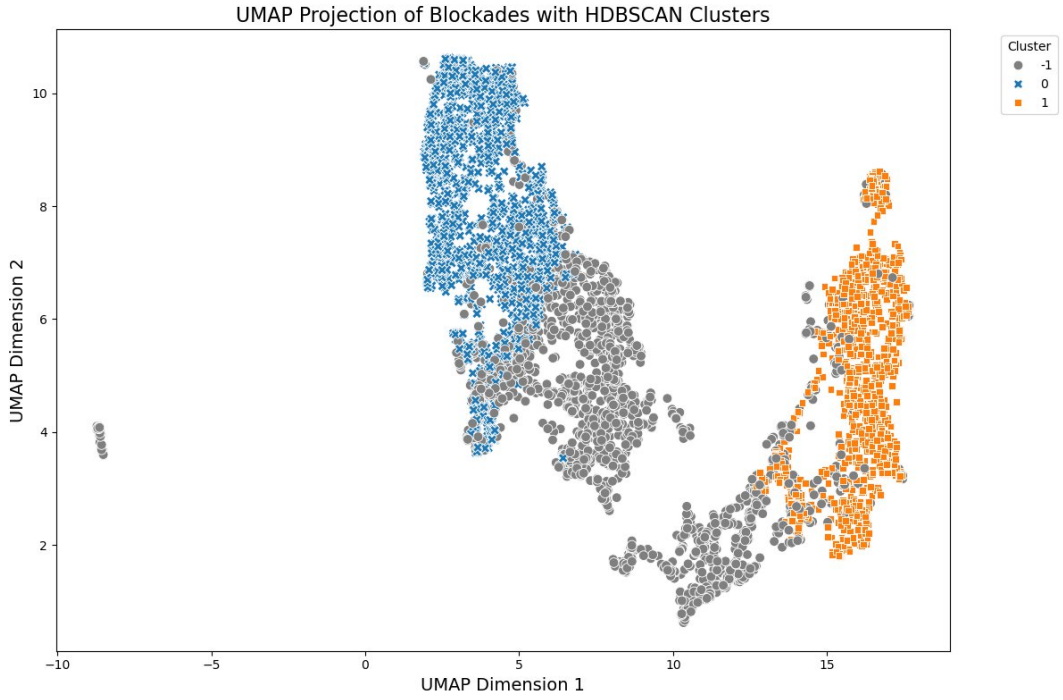
Main Advantage:

Automatically selects important features and discards irrelevant ones by setting their coefficients to zero.

Early Clustering Results on K8-K12 Acetylation Datasets

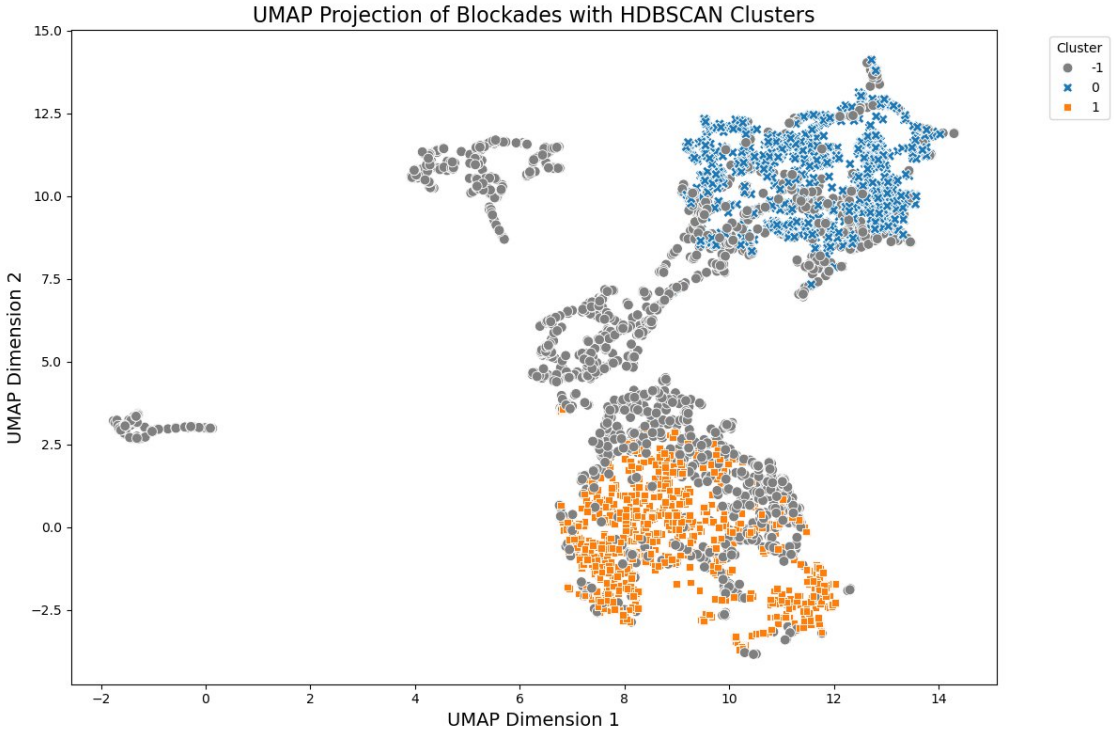


With mean-related features

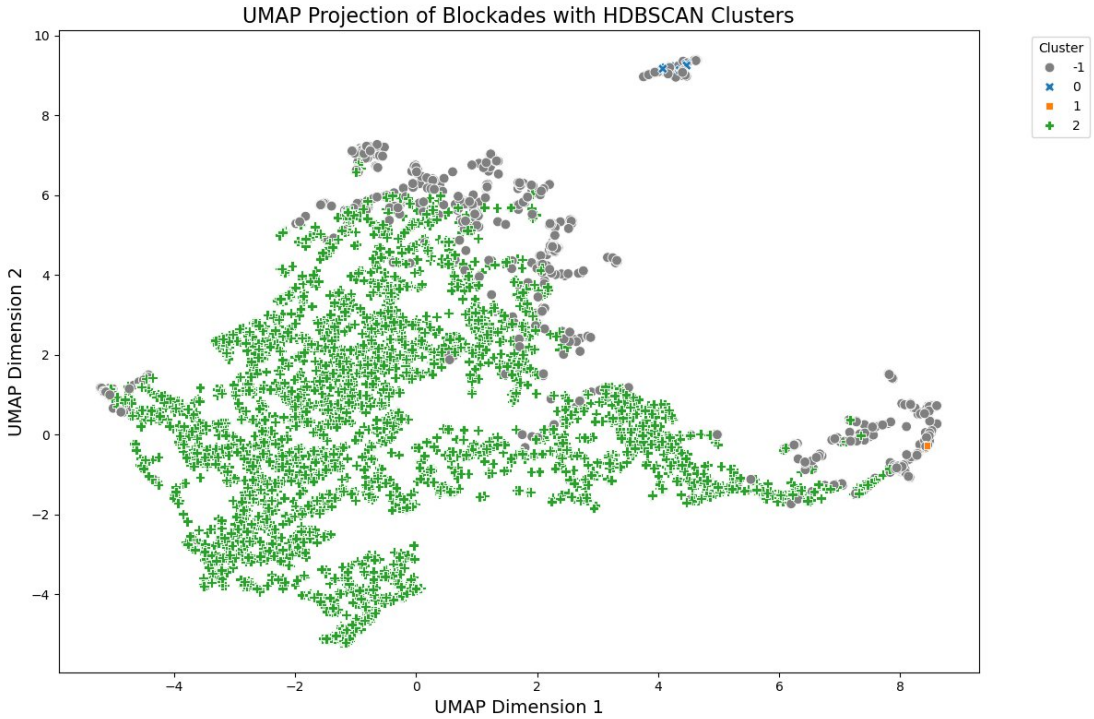


Without mean-related features

Early Clustering Results on K8-K12 Acetylation Datasets



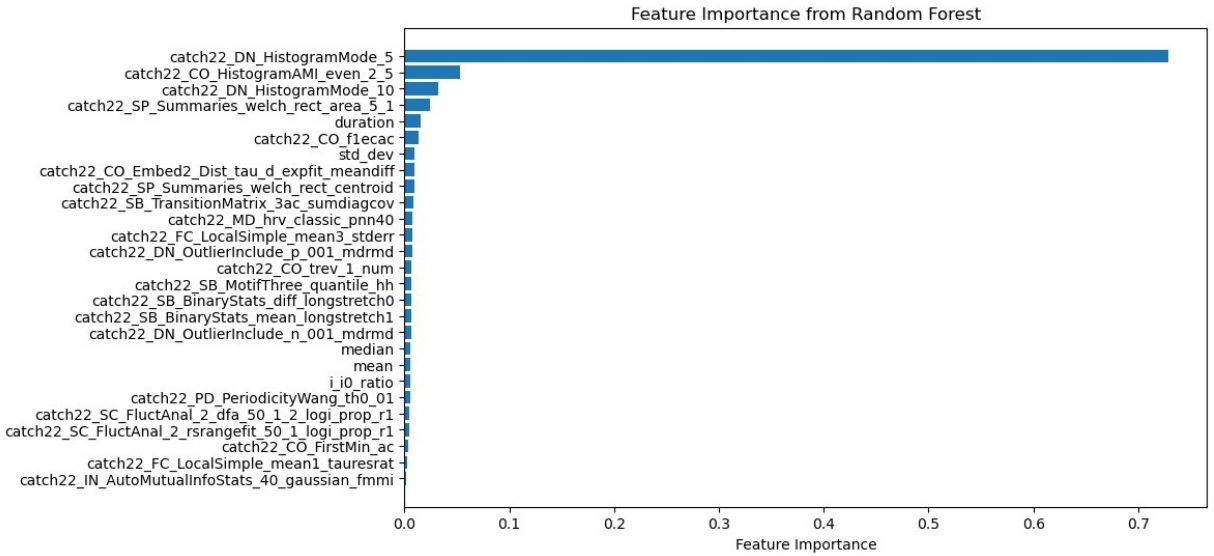
With mean-related features

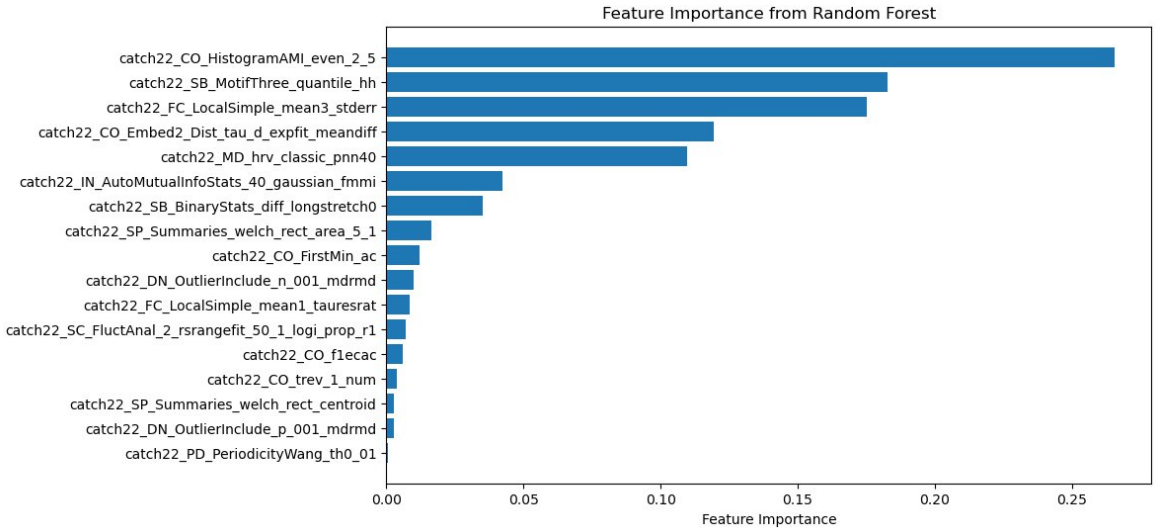


Without mean-related features

Feature Sets

- An overview of how mean-related features can influence models (based on Catch-22 dataset).
- [With and without mean-related features]





Feature Sets

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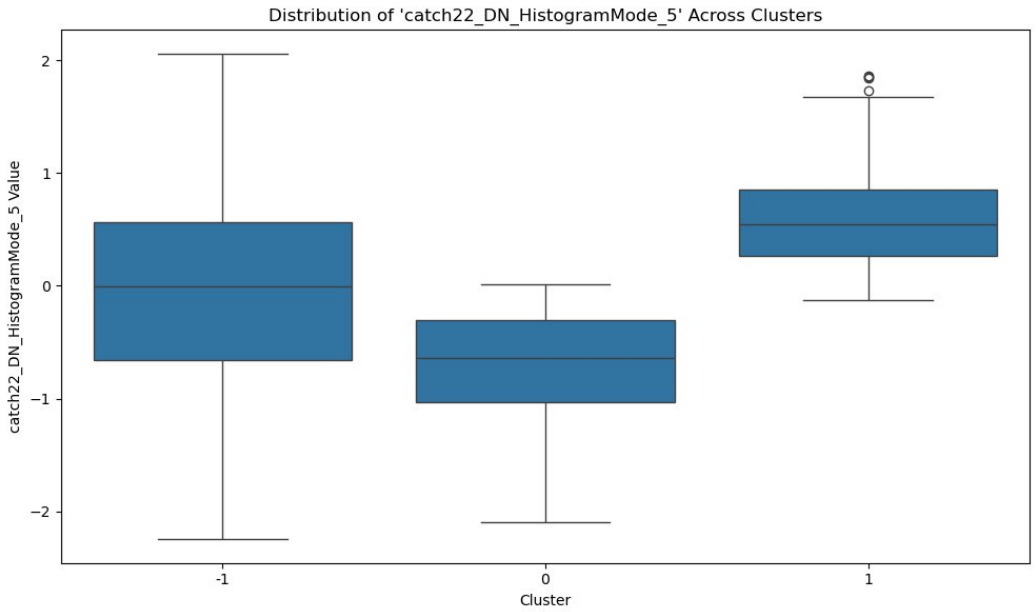

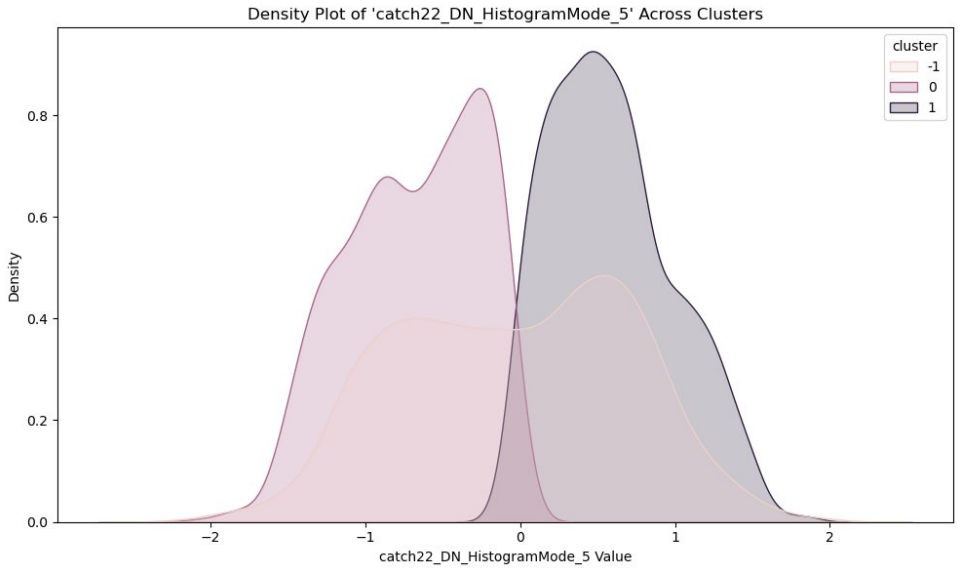

Feature Sets

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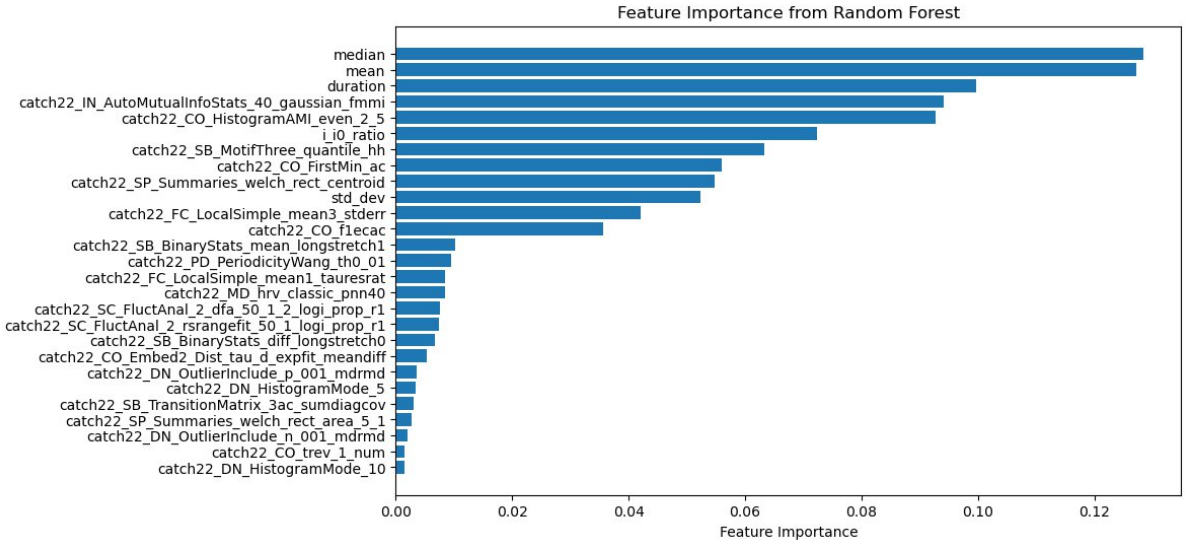
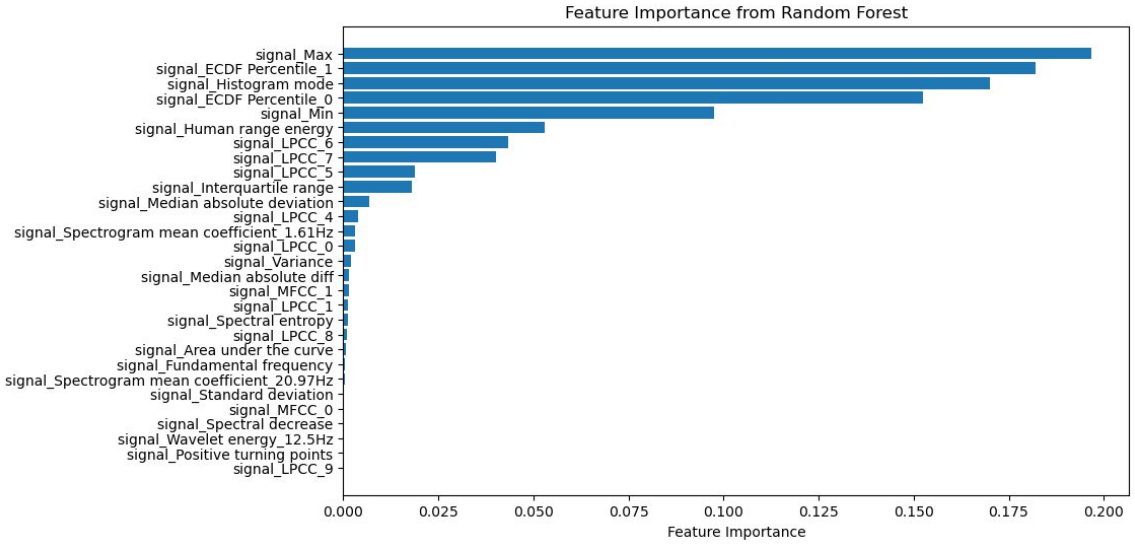
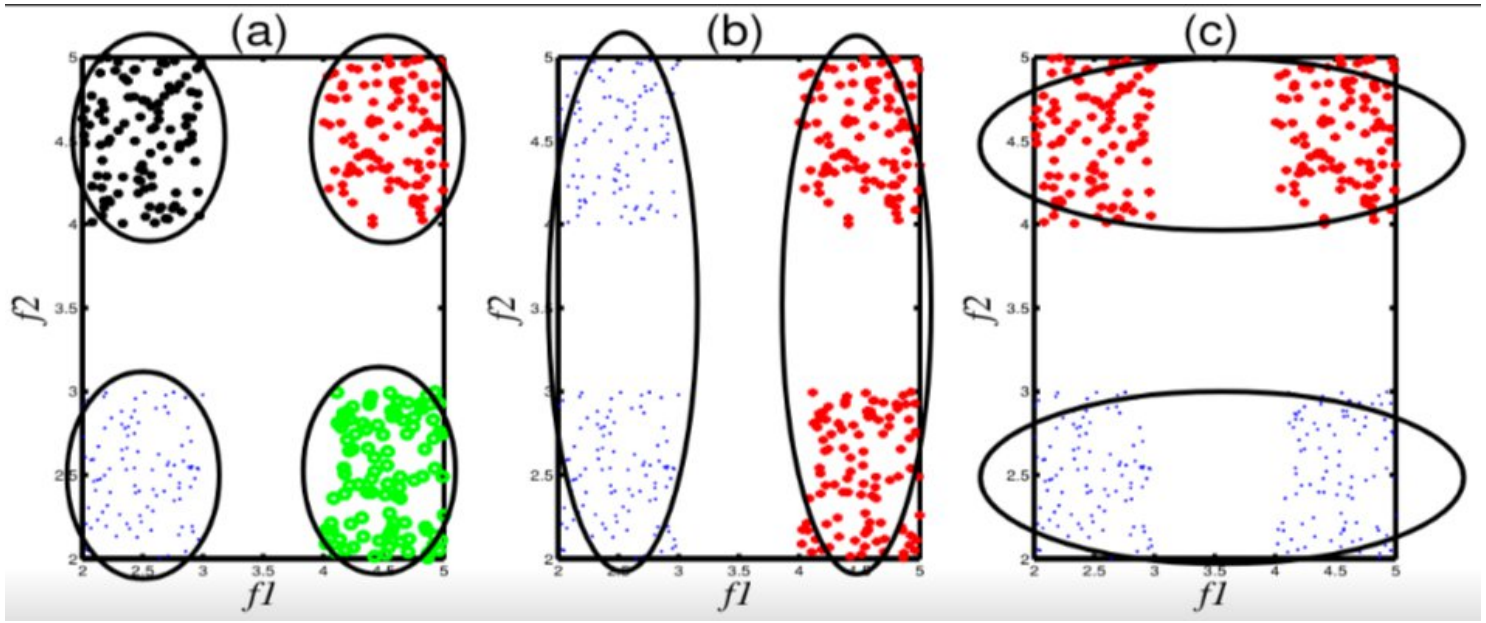




- Clusters depend on the feature sets



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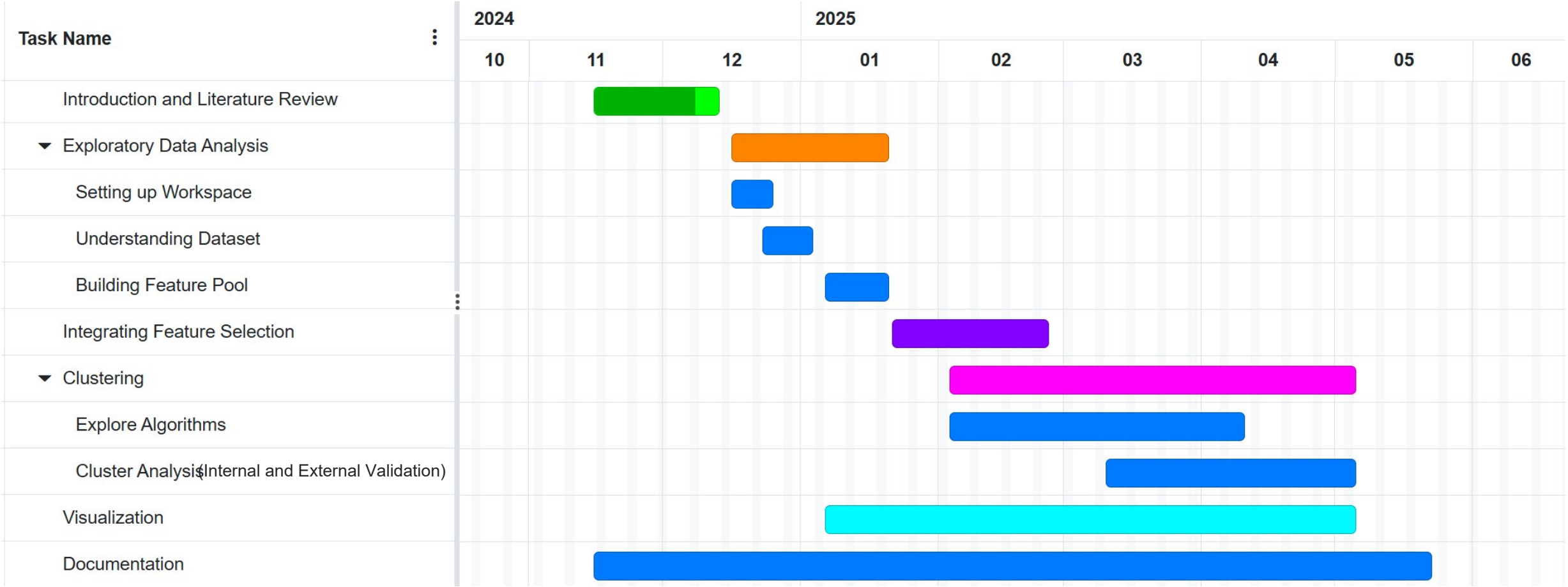
Work Packages Status

- **Literature Review:**
- **Exploratory Data Analysis:**
- **Integrated Feature Selection and Clustering:**
  - Integrate the feature selection into clustering so it is automatically performed
  - Use the feature pool from the previous work package
- **Cluster Analysis and Interpretation:**
  - Use context from known data about the analytes
  - Conduct temporal analysis when analytes are added dynamically
  - Outlier detection
- **Visualization**

Timeline Update



Outlook

- Catch-22 dataset did not offer a full range of features, as on the H4 acetylation dataset
- We continue with using the TSFEL dataset for further analysis
- Lasso regularization as described in Witten et al. provides a useful technique for feature selection, which would help discard redundant features

References

[1] Witten, Daniela M., and Robert Tibshirani. "A framework for feature selection in clustering." Journal of the American Statistical Association 105.490 (2010): 713-726.

[2] Aggarwal, C.C., & Reddy, C.K. (Eds.). (2014). Data Clustering: Algorithms and Applications (1st ed.). Chapman and Hall/CRC. https://doi.org/10.1201/9781315373515

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Thank You